SOFTWARE REQUIREMENTS SPECIFICATION(SRS)

FOR

PERMISSION TO OCCUPY (PTO) MANAGEMENT SYSTEM

Developed For: Mangaung Metropolitan Municipality

Developed by: Group D

Student Initials and Surname	Student Number	Role
A Mnyamanzi	224113476	UI/UX Design
B Hlungwane	224022767	Project Manager
BS Luvhengo	224015143	Technical Lead
BN Mothudi	224124772	Requirements Lead
M Mlambo	223059218	Database Lead
U Hangwani	224073925	CEO/Founder/Testing Lead

Version 2

Module: ITC327W – Work-Integrated Learning

Programme: Diploma in Information Technology

Assessment: Phase 1 & Phase 2

Lecturer: Ms M Mbele

Date: 12 October 2026

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Phase 1

1.Project Introduction

The proposed project is a **Permission To Occupy** (PTO) Management System for the Mangaung Metropolitan Municipality's Human Settlements Department.

The system will provide a centralized digital platform for managing PTO records and will support functions such as user management, beneficiary management, site management, PTO record management, status monitoring, expiry monitoring, renewal, de-registration, searching, filtering and reporting. The system is intended to address the limitations associated with manual and spreadsheet-based record management while providing controlled access to information according to the user's role.

The solution will consist of a Flutter mobile application, an ASP.NET Core web application, and a shared Supabase database backend.

The main functions of the system will include beneficiary management, site management, PTO creation, expiry monitoring, PTO renewal, PTO de-registration, searching, filtering, dashboard monitoring and data security.

Project statement:

The organization currently uses Excel spreadsheets and manual processes to complete the capturing and managing of PTO. This results in difficulty in managing large numbers of beneficiaries, difficulty monitoring PTO valid periods and difficulty maintaining accurate information about sites and beneficiaries. The problem affects municipal officials and PTO beneficiaries because officials may struggle to efficiently track and manage beneficiary and PTO information, while beneficiaries may not receive timely reminders about the expiry or renewal of their PTOs. An improved solution is needed to digitize and centralize the PTO management process, including capturing, monitoring, tracking and managing PTO records.

The proposed PTO Management System will provide a centralized digital platform for authorized municipal users to capture and manage beneficiary, site and PTO information. A System Administrator will manage authorized users, while the PTO Officer/User will capture and manage beneficiary, site and PTO information. Beneficiaries will have controlled access to view their own PTO-related information and status.

The proposed solution is intended to reduce reliance on manual and spreadsheet-based processes, improve the accessibility and organization of PTO information, support the monitoring of PTO records and provide role-based access to information. The system will be developed as an academic prototype based on the requirements and information confirmed by the stakeholder.

2.Stakeholder and Problem Analysis

2.1 Stakeholder, Organisation and User Profile

The proposed PTO Management System is intended to support the Mangaung Metropolitan Municipality's management of Permission to Occupy records associated with informal settlements. The system aims to improve accessibility, efficiency and transparency by allowing beneficiaries to apply for PTOs independently while aiding those who may experience difficulties accessing or using the system.

- The **System Administrator** will be responsible for managing authorised system users, including PTO Officers, managing available sites by adding or removing site information, and viewing and generating reports. The administrator will ensure that the system is properly managed and that authorised users have appropriate access to its functionality.

- **PTO Applicants:** Our system consists of 2 Users, the **Beneficiaries and PTO Officers.**

    - **Beneficiary:** The Beneficiary will be the primary applicant using the system. Beneficiaries will be able to apply for PTOs, log in, upload the required supporting documents, track and view their PTO applications and statuses, receive notifications and update their personal information. They will be able to complete the application process independently, provided they have access to a suitable device and the necessary digital skills.

    - **PTO Officer:** The PTO Officer will be responsible for assisting beneficiaries who do not have access to suitable electronic devices or lack the necessary digital skills to apply independently. PTO Officers may be deployed to communities to assist beneficiaries with completing PTO applications, uploading required documents and capturing the necessary information on their behalf. Their role will primarily focus on providing application assistance and ensuring that beneficiaries who face technological barriers can access the PTO application process.

2.2 Current Process

The current PTO process at Mangaung Metropolitan Municipality is largely manual. Forms are completed manually by the relevant parties, and the information is subsequently captured and maintained using spreadsheets.

The process relies on employees to complete forms correctly, capture information accurately and maintain the relevant spreadsheet records. Information therefore must pass through several manual activities before it can be processed and stored.

Based on the latest stakeholder engagement, the PTO process occurs within the broader context of informal settlements. Individuals may initially settle informally on available land, after which the settlement may go through processes of recognition and formalisation by the municipality. As part of the formalisation process, sites are identified and measured according to the applicable municipal requirements. Once the relevant sites and beneficiaries have been identified, Permission to Occupy records may be issued.

2.3 Identified Problems

The organisation currently uses **Excel spreadsheets and manual processes** to capture and manage PTO information. This makes it difficult to manage many beneficiaries, monitor PTO validity periods and maintain accurate information about sites and beneficiaries.

The stakeholder stated that there is currently **no dedicated system for PTO management**, and that information is captured using Excel spreadsheets, which can leave the process open to manipulation.

The consequences include:

- Difficulty managing large numbers of beneficiaries.
- Difficulty monitoring six-month PTO validity periods.
- Difficulty keeping beneficiary and site information up to date.
- Increased reliance on manual processes.
- Possible errors or manipulation of information.
- Difficulty identifying PTOs that are approaching expiry

2.4 Affected Users

The problem affects:

- Municipal officials as they need to manually capture and manage PTO information.

- PTO beneficiaries because delays to receive the PTO, or apply for it and they are no system to remind or communicate

2.5 Problem Statement

The organisation currently uses Excel spreadsheets and manual processes to complete the capturing and managing of PTOs. This results in difficulty in managing large numbers of beneficiaries, monitoring PTO validity periods and maintaining accurate information about sites and beneficiaries. The problem affects municipal officials and PTO beneficiaries because officials may struggle to efficiently track and manage PTO information, while beneficiaries may not receive timely reminders regarding the expiry or renewal of their PTOs. An improved solution is needed to digitise and centralise the PTO management process, including capturing, monitoring, tracking and managing PTO records.

3.Project Aim and Objectives

Purpose

The purpose of this project is to develop a digital system that assists the municipality with managing and monitoring Permissions to Occupy.

3.1 Aim

To improve the efficiency, accuracy and monitoring of the PTO management process through a mobile and web-based system.

3.2 Objectives

The objectives of the project are to:

- Capture beneficiary information digitally.
- Capture and manage site information.
- Create and manage PTO records.
- Monitor the six-month PTO validity period.
- Provide reminders when a PTO is approaching expiry.
- Allow authorised officials to update and manage PTO statuses.
- Allow authorised officials to renew PTOs.
- Allow authorised officials to de-register PTOs.
- Provide search and filtering functionality.
- Provide a dashboard with an overview of important PTO information.
- Reduce dependence on Excel spreadsheets and manual processes.
- Provide centralised storage of PTO information.

- Protect confidential beneficiary information.

4.Project Scope

The project focuses on the design and development of an academic prototype of a Permission to Occupy (PTO) Management System for Mangaung Metropolitan Municipality. The system will provide functionality for authorised municipal users to manage system users, beneficiaries, sites and PTO records, while beneficiaries will have controlled access to view their PTO-related information. The system will support PTO monitoring, status management, expiry monitoring, renewal, de-registration, searching, filtering and reporting where required.

The system will focus specifically on managing PTO information and will not attempt to manage the complete Human Settlements process.

4.1 Deliverables

The project will deliver:

- A **Flutter mobile application** for accessing and managing PTO-related information.
- An **ASP.NET Core web application** for municipal officials to manage the PTO process.
- A **Supabase database** to store beneficiary, site and PTO information.
- A **beneficiary management function** to capture and manage beneficiary details.
- A **site management function** to capture and manage site information.
- A **PTO management function** to create, update and track PTO records.
- A **PTO validity monitoring function** to track the six-month validity period.
- A **notification or reminder function** to alert beneficiaries or officials when a PTO is approaching expiry.
- A **search function** to locate beneficiary, site and PTO records.

User authentication and access control to protect sensitive information

4.2 Acceptance Criteria

The system will be accepted if:

- Municipal officials can successfully capture and manage beneficiary information.
- Site information can be captured and linked to the relevant beneficiary.
- PTO records can be created, updated and tracked.
- The system can identify PTOs that are approaching the end of their six-month validity period.
- Users can search for beneficiary, site and PTO information.
- Authorised users can securely access the system.
- Sensitive beneficiary information is protected from unauthorised access.
- Information is stored successfully in the centralised database.
- The mobile and web applications perform the required PTO management functions.

4.3 Exclusions

The following are outside the scope of the project:

- Management of the complete housing or human settlements process.
- Management of RDP housing applications and subsidies.
- Issuing or registering title deeds.
- Complete township or informal settlement formalisation.
- Replacing existing systems such as the National Housing Needs Register or HSS Online.
- Automatic approval of beneficiaries for housing subsidies.
- Full integration with all existing municipal or government systems during the initial project.

4.4 Constraints

The project may be affected by the following constraints:

- **Limited funding:** The development team has limited financial resources for hosting, servers and paid software.
- **Time limitations:** The project must be completed within the academic project timeframe.

- **Access to information:** Some requirements and processes may depend on information from municipal stakeholders.
- **Municipal processes:** Internal approval, procurement and technology processes may affect future implementation.
- **Integration limitations:** Access to existing government systems may not be available during the development of the initial system.
- **Stakeholder availability**: Access to detailed municipal PTO processes may be limited because relevant municipal personnel are involved in other operational programmes and activities.
- **External system uncertainty**: A similar PTO-related programme may be under development by the National Department of Human Settlements. The academic prototype will therefore not assume integration with or replacement of such a programme.

4.5 Assumptions

The project assumes that:

- The project assumes that the requirements confirmed by the stakeholder will be sufficient to develop the academic prototype.
- Users will have access to a smartphone or computer and an internet connection.
- Beneficiary and site information will be available for capturing into the system.
- Municipal officials will use the system to manage and monitor PTO records.
- The proposed technologies will be available throughout the development period.
- Any PTO requirements, business rules or procedures that have not been confirmed by the stakeholder will be treated as TBD and will not be assumed to represent official municipal policy.

5.Stakeholder Needs

The stakeholder needs for the proposed PTO system were identified based on the roles of the stakeholders and the challenges experienced in the existing manual process. The system should address these needs while improving efficiency, transparency, accountability and the accuracy of PTO records.

❖ Internal Departments

Internal departments are the primary users of the proposed system and therefore have the greatest need for functionality that supports the day-to-day PTO process.

Their main needs include:

- Centralised information: A single system where PTO applications and related information can be captured, stored and accessed by authorised users.
- Efficient processing: A system that reduces the time required to complete and process PTO applications compared with the current manual process.
- Accurate data: Validation and structured data capturing to reduce errors caused by manual data entry.
- Application tracking: The ability to view and track the status of PTO applications throughout the process.
- Controlled access: Role-based access to ensure that users can only access or modify information relevant to their responsibilities.
- Approval workflow: A structured process for reviewing and approving or rejecting applications.
- Audit trail: The ability to record important actions performed by users to improve accountability and reduce opportunities for unauthorised manipulation.
- Search and reporting: The ability to search for records and generate relevant information or reports when required.
- Secure record keeping: Protection of PTO information against unauthorised access, loss or unauthorised changes.

❖ Mangaung Greater Community

The greater Mangaung community is an indirect stakeholder. Their needs are primarily related to the quality, fairness and reliability of municipal service delivery.

Their main needs include:

- Improved service delivery: Municipal processes should be handled efficiently and within reasonable timeframes.
- Transparency: Processes should be sufficiently controlled to reduce opportunities for manipulation or unfair treatment.
- Accountability: There should be clear responsibility for actions performed during the PTO process.

- Reduced delays: Improved internal processing should contribute to fewer delays that may affect service delivery.
- Reliable information: Information used during the process should be accurate and properly maintained.

6. Functional requirements

FR1: User Authentication

The system shall allow authorised users to securely log in according to their assigned role.

FR2: User Management

The system shall allow the System Administrator to create, register, assign and manage authorised system users.

FR3: Beneficiary Management

The system shall allow authorised PTO Officers/Users to capture, view and update beneficiary information.

FR4: Site Management

The system shall allow authorised Administrators to capture and manage site information, including site number, location and site size where applicable.

FR5: PTO Creation

The system shall allow authorised officials to create a PTO record for an eligible beneficiary.

FR6: PTO Status

The system shall allow officials to view and update the status of a PTO.

Possible statuses can include:

- Pending
- Approved
- Active
- Expiring
- Expired
- Renewed
- De-registered

FR7 – Beneficiary PTO Viewing

The system shall allow beneficiaries to securely view their PTO-related information, including relevant beneficiary, site and PTO status information.

FR8: PTO Expiry Monitoring

The system shall calculate and monitor the six-month PTO validity period.

FR9: Expiry Notifications

The system shall provide reminders when a PTO is approaching expiry.

FR10 – PTO Renewal

The system shall allow authorised PTO Users to renew an existing PTO where renewal is required. The renewed PTO shall have an updated validity period and status.

FR11 – PTO De-registration

The system shall allow authorised PTO Users to de-register a PTO and record the reason and date.

FR12: Search

Authorised officials shall be able to search for beneficiaries, sites and PTO records.

FR13: Filtering System

The system shall allow authorised officials to filter beneficiaries, sites and PTO records using relevant criteria, such as:

- PTO status
- Expiry date
- Informal settlement or location
- Site number
- Beneficiary details

FR14: Dashboard

The system shall provide a dashboard that gives authorised officials an overview of PTO information. The dashboard shall display relevant information such as:

- Total number of beneficiaries
- Total number of sites

- Active PTOs
- PTOs approaching expiry
- Expired PTOs
- Renewed PTOs
- De-registered PTOs

FR15: Record History

The system shall maintain a history of changes made to PTO records.

FR16: Role-Based Access Control

The system shall restrict functionality and information according to the user's assigned role.

7. Non- Functional Requirements

7.1 Security

- User authentication must be implemented.
- Authorisation must restrict sensitive functions.
- Personal information must be protected.

7.2 Performance

- The system should respond efficiently to normal user requests.
- Searches should return results without unnecessary delays.
- The system should be capable of handling significant number of beneficiary and site records

7.3 Usability

- The interface should be simple and easy for municipal officials to use.
- The mobile application should be suitable for Android/iOS environments.

7.4 Reliability

- Data should not be lost during normal operation.
- The system should maintain consistent records.

7.5 Scalability

The system should allow additional informal settlements, beneficiaries and sites to be added.

7.6 Maintainability

The system should use a modular architecture to allow future development and maintenance.

7.7 Availability

The system should be accessible online to authorised users where an internet connection is available.

8. Integration Requirements

The **Flutter mobile application, ASP.NET Core and Supabase database backend** shall operate together as one integrated PTO Management System.

IR1 — Flutter and ASP.NET Core Integration

The Flutter mobile application shall communicate with the ASP.NET Core backend through APIs to send and retrieve PTO-related information.

IR2 — ASP.NET Core and Supabase Integration

The ASP.NET Core backend shall communicate with the Supabase database to store, retrieve, update and manage beneficiary, site and PTO information.

IR3 — Data Flow Integration

Information captured through the Flutter mobile application or web application shall be processed by ASP.NET Core and stored in the Supabase database.

IR4 — Centralised Data

Both the Flutter mobile application and ASP.NET web application shall use the same centralised Supabase database to ensure that authorised users access consistent and up-to-date information.

IR5 — Authentication Integration

User authentication and authorisation shall be integrated across the Flutter application, ASP.NET Core backend and Supabase to ensure that only authorised users can access sensitive PTO information.

IR6 — Real-Time Data Updates

When beneficiary, site or PTO information is created or updated, the changes shall be stored in the central database and made available to authorised users through the integrated applications.

IR7 — Notification Integration

The integrated solution shall support the identification of PTOs approaching their expiry date so that reminders or notifications can be generated for the relevant users.

The proposed solution uses **Flutter for the mobile application, ASP.NET Core for the web application/backend, and Supabase for storing data**, as identified by the development team during the stakeholder engagement.

9. Feasibility Study: PTO Management System

9.1 Technical Feasibility

- **Flutter** for the mobile application.
- **ASP.NET Core/C#** for the web application and backend.
- **Supabase** for storing and managing beneficiary, site and PTO data.
- **GitHub** for source-code management and collaboration.

9.2 Operational Feasibility

- The proposed system will allow officials to:
- Capture beneficiary information.
- Manage site information.
- Create and manage PTO records.
- Monitor six-month PTO validity periods.
- Renew and de-register PTOs.
- Search and filter records.
- View important information through a dashboard.
- Receive reminders about PTOs approaching expiry.

The system is suitable for municipal officials because it centralises information and reduces reliance on manual spreadsheets.

A possible limitation is that users may require training to use the new system effectively. The system should therefore have a simple and user-friendly interface.

9.3 Economic Feasibility

The project is economically feasible as a student prototype because the group plans to use cost-effective and accessible technologies.

The main resources required include:

- Computers or laptops.
- Internet access.

- Flutter and development tools.
- ASP.NET Core development tools.
- Supabase database services.
- GitHub for version control.

9.4 Schedule Feasibility

The project must be completed within the available academic semester.

9.5 Legal Feasibility

The system will store ID numbers, contact details and other confidential information. This means that the system will have to restrict access to authorised users, protect sensitive information and avoid disclosing personal information. The stakeholder specifically indicated that ID numbers and similar personal information should not be disclosed.

9.6 Ethical Feasibility

The system must use beneficiary information responsibly and fairly.

Ethical considerations include:

- Protecting beneficiary privacy.
- Preventing unauthorised access to personal information.
- Ensuring that PTO information is not manipulated.
- Treating all beneficiaries fairly.
- Ensuring that authorised officials are accountable for changes made to records.
- Avoiding discrimination or unfair treatment when managing beneficiaries.
- Keeping accurate records.

The development team must also ensure that stakeholder information is used only for the purposes of the project and that confidential information is not unnecessarily shared.

10. Project Risk Analysis Summary

The project risk analysis identified 12 potential risks that may affect the successful development and future implementation of the proposed PTO

system. The risks were assessed using a **Risk Score = Likelihood × Impact approach**. Likelihood and impact are each rated on a scale from 1 to 3, producing risk scores between 1 and 9.

10.1 Risk levels:

Risk Score	Risk Level	Interpretation
1-3	Low	The risk has a relatively small effect on the project and can be monitored.
4-6	Medium	The risk could have a noticeable effect and requires mitigation.
7-9	High	The risk could seriously affect project completion or quality and requires immediate attention.

10.2 Overall Risk Assessment

The project has been assessed as having an overall HIGH risk level. The register contains:

2 High risks

8 Medium risks

2 Low risks

The two highest risks, R01 and R02, both have a score of 9, meaning they are Likely to occur and could have a high impact on the project.

10.3 Key Risks Identified

R01 – Delayed or Incomplete Stakeholder Requirements (Score: 9 – High)

There is a significant risk that the required PTO criteria, documentation, business rules and other system requirements may not be received from the stakeholder on time. This could result in incomplete or incorrect requirements and affect the system design and development.

The project team should maintain regular communication with the stakeholder; document all confirmed requirements and clearly identify outstanding requirements as TBD. Development should continue using the confirmed requirements while awaiting the outstanding information.

R02 – Municipal Approval and Internal Processes (Score: 9 – High)

Municipal approval procedures may delay the confirmation of requirements, access to information or eventual adoption of the system. Since the organisation is a municipality, internal governance, licensing, security and approval requirements may need to be satisfied before the system can be deployed.

The team should identify the relevant municipal stakeholders early and treat municipal deployment as a future phase, while continuing development of the academic prototype.

R11 – Failure to Complete the Project Within the Academic Deadline (Score: 6 – Medium)

There is a possibility that the team may not complete all planned functionality within the academic Phase One deadline. This risk is increased by the possibility of outstanding stakeholder information being received late.

The recommended response is to prioritise the Minimum Viable Product (MVP) and core PTO functionality, allocate tasks among team members and defer non-essential functionality to a later phase.

10.4 Other Medium Risks

Several other risks received a score of 4 (Medium). These include:

R03 – Data and security: Unauthorised access to sensitive beneficiary information.

R04 – Limited project funding: Restrictions on paid hosting, SMS services and infrastructure.

R05 – Understanding the current manual process: Difficulty translating the existing spreadsheet/manual process into accurate system requirements.

R06 – PTO validity and expiry: Incorrect implementation of the six-month PTO validity and notification process.

R07 – SMS integration: SMS services may be unavailable, unaffordable or technically incompatible.

R08 – Future Supabase Environment Requirements (Score: 2 – Low) and R12 – Future Migration to Municipal Infrastructure (Score: 2 – Low) were classified as low risks because they are considered unlikely to occur during the academic development phase. However, they may become more significant if the municipality decides to adopt the system in the future.

R09 – System scalability: The system may experience difficulties when handling large numbers of beneficiaries and sites.

R10 – Incorrect data entry: Incorrect beneficiary, site or PTO information could affect monitoring and administrative decisions.

10.5 Risk Analysis Conclusion

Overall, the PTO system project is classified as HIGH risk, primarily because of uncertainties surrounding stakeholder requirements and municipal approval processes. These risks could affect the project's requirements, system design, development and potential future deployment.

The project team should therefore focus immediately on obtaining and confirming outstanding stakeholder requirements, identifying the necessary municipal stakeholders and approvals, and prioritising the core functionality required for the academic MVP. Medium and low risks should continue to be monitored throughout the project lifecycle.

In summary, the project's risk management strategy is to address the two high-priority risks immediately, reduce medium-level risks through preventative measures, and monitor low-level risks while maintaining flexibility for future municipal deployment.

11. Requirements change log

Change ID	Original Requirement	Change Made	Reason for Change
RC1	The project initially considered a broader Human Settlements system.	The scope was changed to focus specifically on **PTO Management**.	A complete Human Settlements system would be too large and complex to develop within the available semester.
RC2	Integration with existing government systems was considered.	Integration with systems such as the National Housing Needs Register and HSS Online was removed from the initial prototype.	Access to these systems and their integration requirements may not be available to the group.
RC3	The system focused on capturing and	**PTO expiry monitoring** was added.	PTOs have a validity period and officials need

Change ID	Original Requirement	Change Made	Reason for Change
	managing PTO records.		to monitor when they are approaching expiry.
RC4	PTO renewal was not initially included.	**PTO renewal** was added as a functional requirement.	Officials need to manage PTOs that need to be renewed.
RC5	PTO de-registration was not initially included.	**PTO de-registration** was added as a functional requirement.	Officials need to manage PTOs that are no longer valid or where a beneficiary no longer occupies a site.
RC6	The system did not initially include filtering.	A **filtering system** was added.	Filtering will make it easier to manage and find records when dealing with many beneficiaries and sites.
RC7	The system did not initially include a dashboard.	A **dashboard** was added.	Officials need a quick overview of beneficiaries, sites, active PTOs, expired PTOs and PTOs approaching expiry.
RC8	Security requirements were not clearly defined.	**Access control and protection of confidential information** were strengthened.	The system will store confidential information such as ID numbers and contact details.
RC9	Full deployment to municipal infrastructure was considered as part of the project.	Municipal deployment was changed to a **future consideration**.	The group may not have access to municipal infrastructure, and deployment could involve additional technical, legal and administrative requirements.

Change ID	Original Requirement	Change Made	Reason for Change
RC10	The project could have included many additional functions.	The requirements were prioritised into **Must Have, Should Have and Could Have** categories.	The Unit guide recommends prioritising requirements and focusing on the main problem to maintain a realistic project scope.
RC11	The system was initially described primarily around internal municipal departments and beneficiaries.	The user structure was clarified into three roles: System Administrator, User/PTO Officer and Beneficiary.	The stakeholder clarified the intended system structure during the latest stakeholder meeting.
RC12	The beneficiary was initially considered as a user who could apply for a PTO.	The beneficiary's role was clarified primarily as a viewing user. Authorised users/PTO Officers will capture beneficiary and site information and manage PTO records.	The stakeholder clarified that users should capture beneficiary and site information and that beneficiaries should have viewed access to their PTO information.
RC13	The PTO process was not fully understood in relation to informal settlements.	The system scope was clarified to focus on PTO management as part of the broader informal-settlement formalisation process.	The stakeholder explained that PTO is one stage within a longer process and does not represent the complete Human Settlements process.
RC14	Potential future municipal implementation was considered.	The current development will be treated primarily as an academic/desktop prototype. Full municipal deployment will remain a future consideration	The stakeholder explained that internal municipal limitations prevent full operational implementation during the current project.

12. Refined project scope

The project focuses on the design and development of an academic prototype of a Permission to Occupy (PTO) Management System for Mangaung Metropolitan Municipality. The system will provide functionality for authorised municipal users to manage system users, beneficiaries, sites and PTO records, while beneficiaries will have controlled access to view their PTO-related information. The system will support PTO monitoring, status management, expiry monitoring, renewal, de-registration, searching, filtering and reporting where required.

The system will not attempt to manage the complete informal-settlement formalisation or Human Settlements process. Processes such as RDP housing subsidies, title deeds, complete township formalisation and integration with national government systems remain outside the scope of the academic prototype.

Phase 2

13. System Architecture and Design

13.1 System Architecture

User Groups and Application Access

User / Role	Uses Flutter Mobile?	Uses ASP.NET Web?	Main Purpose
User/ Beneficiary	Yes	Yes	Submit PTO applications, log in, track status, receive expiry notifications, and update personal profiles.
User / PTO Officer (Beneficiary assistant)	Yes (Mobile field capturing)	Yes	Assists applicants who are not computer literate to submit PTO applications, log in, track status, receive expiry notifications, and update personal profiles.
System Administrator	No	Yes	Manage site availability (add/remove available sites), and view & file municipal reports

Component Responsibilities

Component	Main Responsibilities
Flutter Mobile Application	Provides interfaces for beneficiaries to apply for PTOs, upload verification files, log in, track status, and receive notifications. Enables PTO officers to capture site/beneficiary data on-site.
ASP.NET Web Application	Administrative portal for municipal System Admins to verify criteria, add/remove site records, process renewals/de-registrations, filter records, and file system reports.
Supabase Authentication	Handles secure user authentication, role assignments (Admin, Beneficiary).
Supabase Database (PostgreSQL)	Centralized relational database storing beneficiary profiles, site details, qualification metrics, PTO statuses (Pending, Active, Expiring, Expired, Renewed, De-registered), 6-month validity timers, and audit logs.
Supabase Storage	Secure blob storage for required supporting documents (Beneficiary Certified ID, dependant Certified ID, children's birth certificates, proof of income).
External Notification Service	Dispatches SMS or email reminder alerts when a PTO record approaches its 6-month validity expiration.

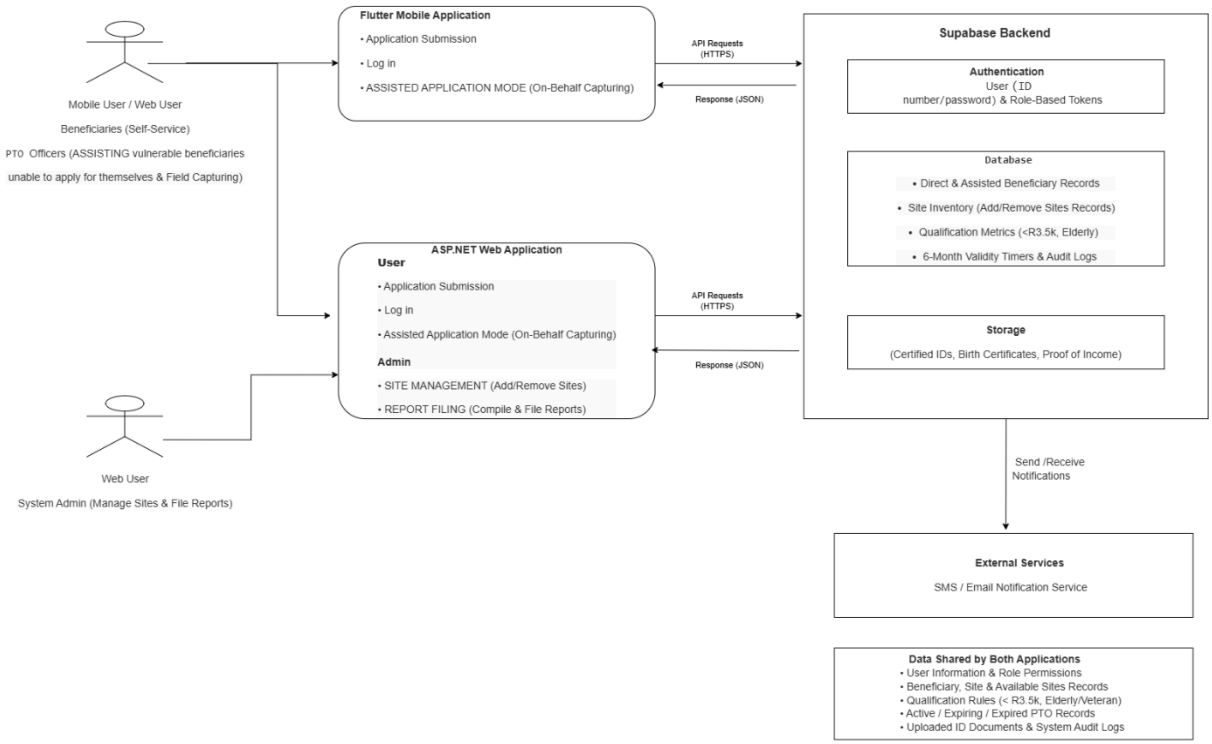
Main Data Flows

From	To	Information Exchanged
Beneficiary / Officer	Flutter Mobile App / ASP.NET Web App	Registration information (income details, and document uploads), login credentials.
Flutter Mobile App / ASP.NET Web App	Supabase Backend	Uploads scanned Certified IDs, birth certificates, and proof of income.
Flutter Mobile App / ASP.NET Web App	Supabase DB / Authentication	Sends login credentials (ID number/password); receives signed JWT token containing user ID and role.

Flutter Mobile App / ASP.NET Web App	Supabase Backend	Queries application s, retrieves attached document URLs, updates PTO statuses (Approved, Renewed, De-registered), and files municipal reports.
Supabase Backend	External Notification Service	Triggers reminder alerts when a 6-month validity period approaches expiry or application status updates.

System Architecture Diagram

https://tinyurl.com/4db6yj2n



Architectural Explanation

1. **Who uses the Flutter application**: Beneficiaries/ PTO Officers use Flutter to, apply for PTOs, log in, track application status, and receive alerts. Mobile field officers also use it to capture informal settlement site data on location.

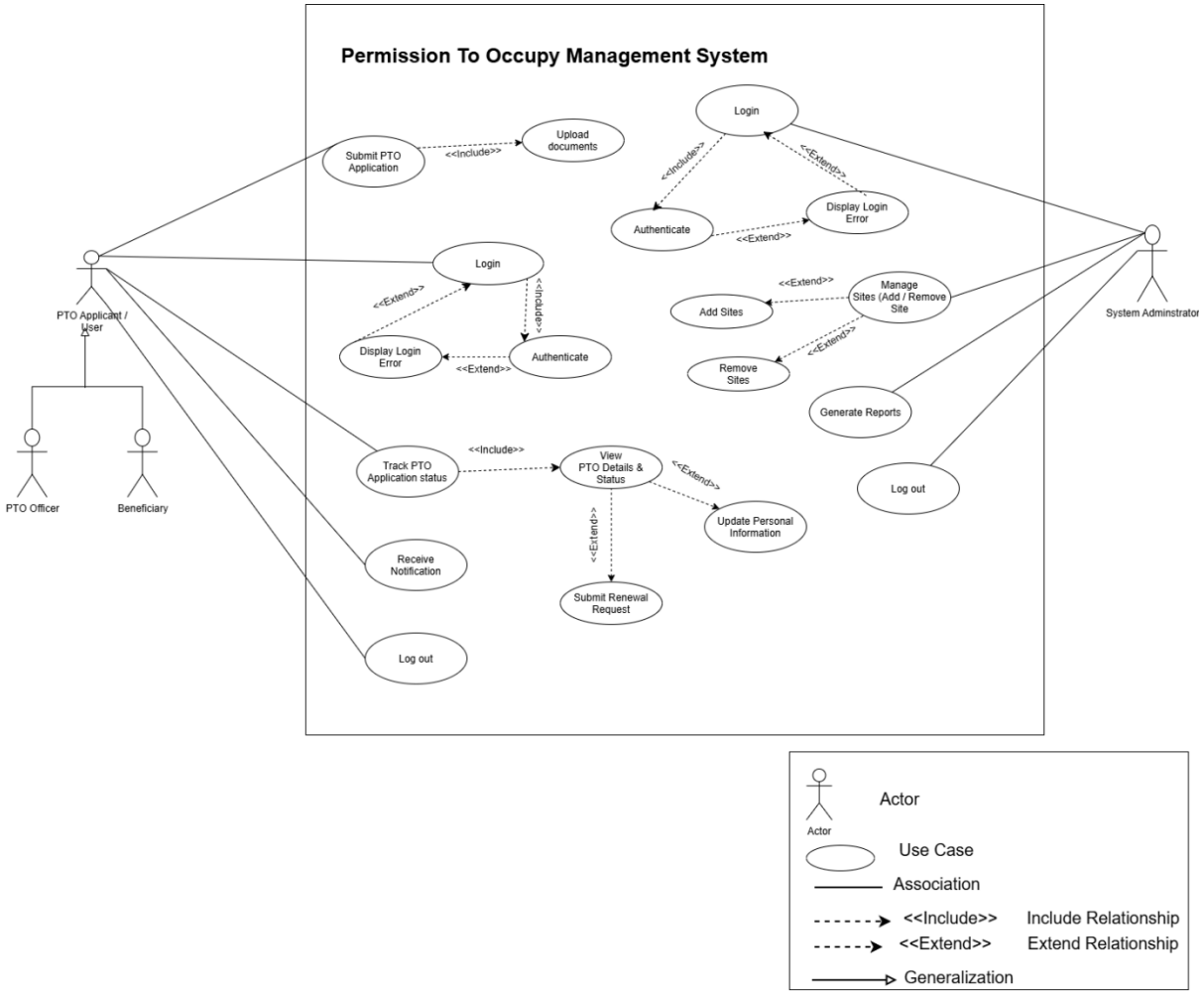
2. **Who uses the ASP.NET application**: Beneficiaries/ PTO Officers use ASP.Net application to apply for PTOs, log in, track application status, and receive alerts. Mobile field officers also use it to capture informal settlement site data on location. System Administrators use it to add or remove available settlement sites and file official reports.
3. **What Supabase is responsible for**: Supabase provides shared authentication, relational database tables (storing beneficiary metrics, site links, PTO timers, and change histories), file blob storage for certified documents, and Row-Level Security (RLS) to safeguard South African ID numbers.
4. **How Flutter communicates with Supabase**: The Flutter app makes HTTPS REST API calls using JSON payloads and JWT tokens to submit applications, stream status changes, and upload files to Supabase Storage.
5. **How ASP.NET communicates with Supabase**: ASP.NET Core connects via the Supabase C# SDK / REST APIs over HTTPS to evaluate applications, update database records, compile dashboard analytics, and export municipal reports.
6. **How the two applications form one integrated solution**: Both client applications share the same central Supabase backend. When a beneficiary applies and documents via the Flutter app, the data is instantly available to System Administrators on the ASP.NET web app for verification, ensuring real-time synchronization across the municipality.

13.2 UML Design

13.2.1 Use Case Diagram

https://tinyurl.com/4ks4tbm2

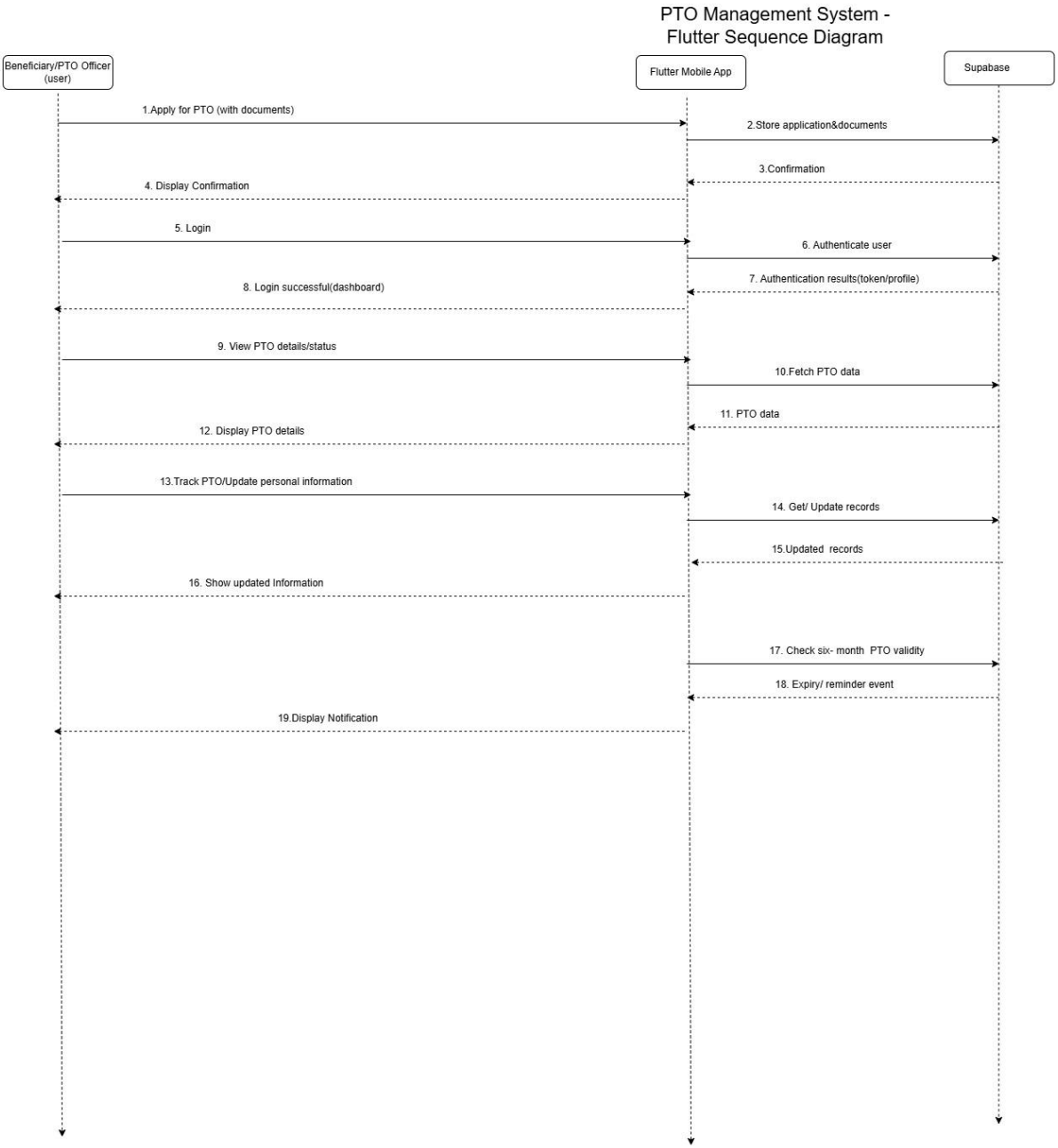
The Use Case Diagram provides an overview of the main functionalities of the PTO Management System and the interactions between users and the system. It identifies three primary actors: the Beneficiary, PTO Officer, and System Administrator. Beneficiaries and PTO Officers can submit PTO applications, upload supporting documents, track application statuses, receive notifications, request renewals, and manage their information. The System Administrator is responsible for managing sites, generating reports, and accessing the system's administrative functions. The diagram also illustrates the relationships between use cases, including include, extend, and generalization relationships.



13.2.2 Flutter Sequence Diagram

https://tinyurl.com/bdcnhxa4

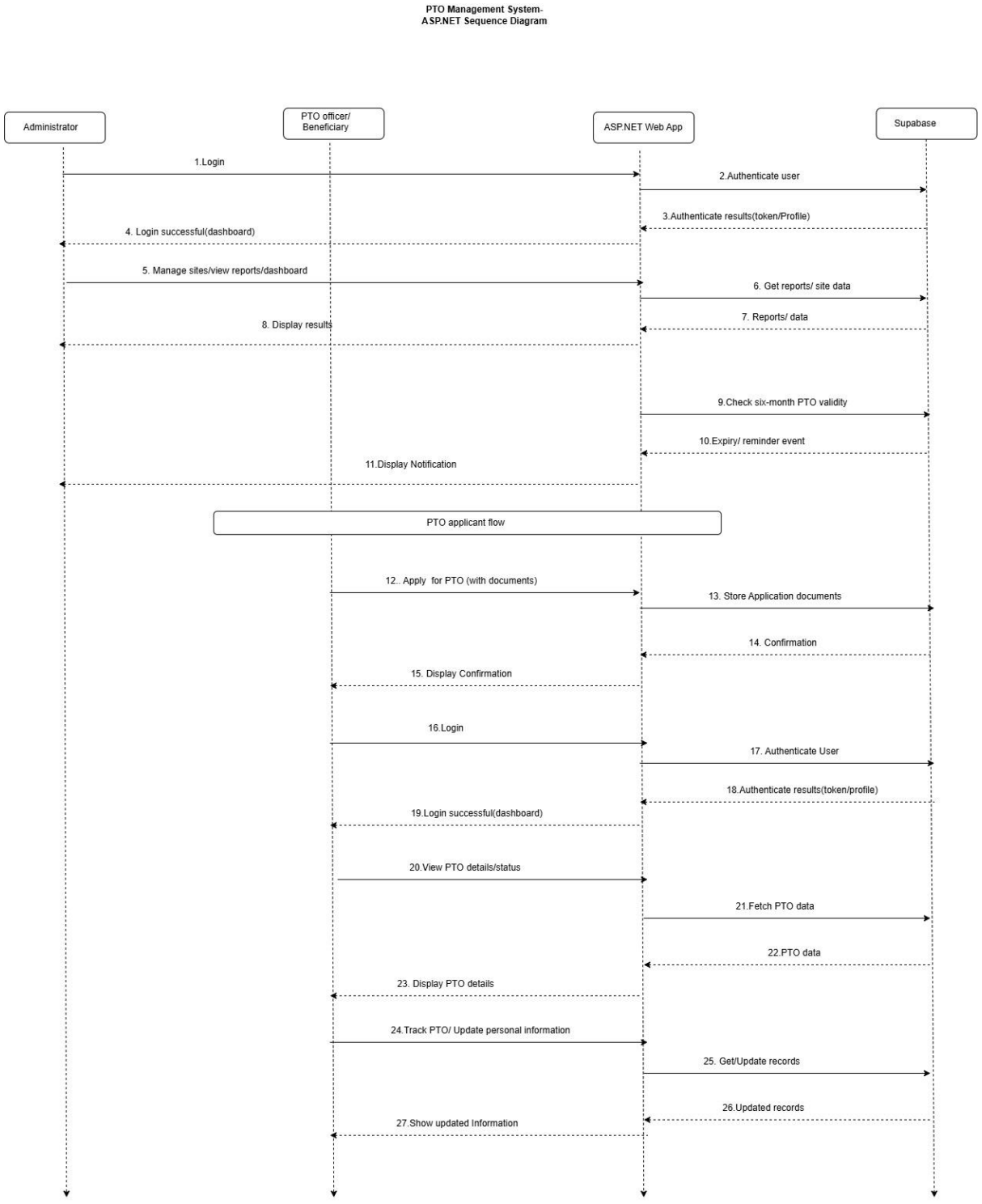
The Flutter Sequence Diagram illustrates the interaction between the Beneficiary/PTO Officer, Flutter Mobile Application, and Supabase database. It demonstrates how users submit PTO applications with supporting documents, log in, view application details and statuses, track PTO records, and update personal information. The diagram also shows how the system checks PTO validity and provides notifications or reminders when PTOs are approaching expiry.



13.2.3 ASP.NET Sequence Diagram

https://tinyurl.com/4ntknas4

The ASP.NET Sequence Diagram illustrates the interaction between the Administrator, PTO Officer/Beneficiary, ASP.NET Web Application, and Supabase database. It demonstrates how administrators log in, manage sites, access reports, and receive PTO expiry notifications. It also shows how PTO applicants submit applications, authenticate, view their application statuses, and update personal information. The diagram highlights how the web application communicates with Supabase to retrieve, store, and update information.

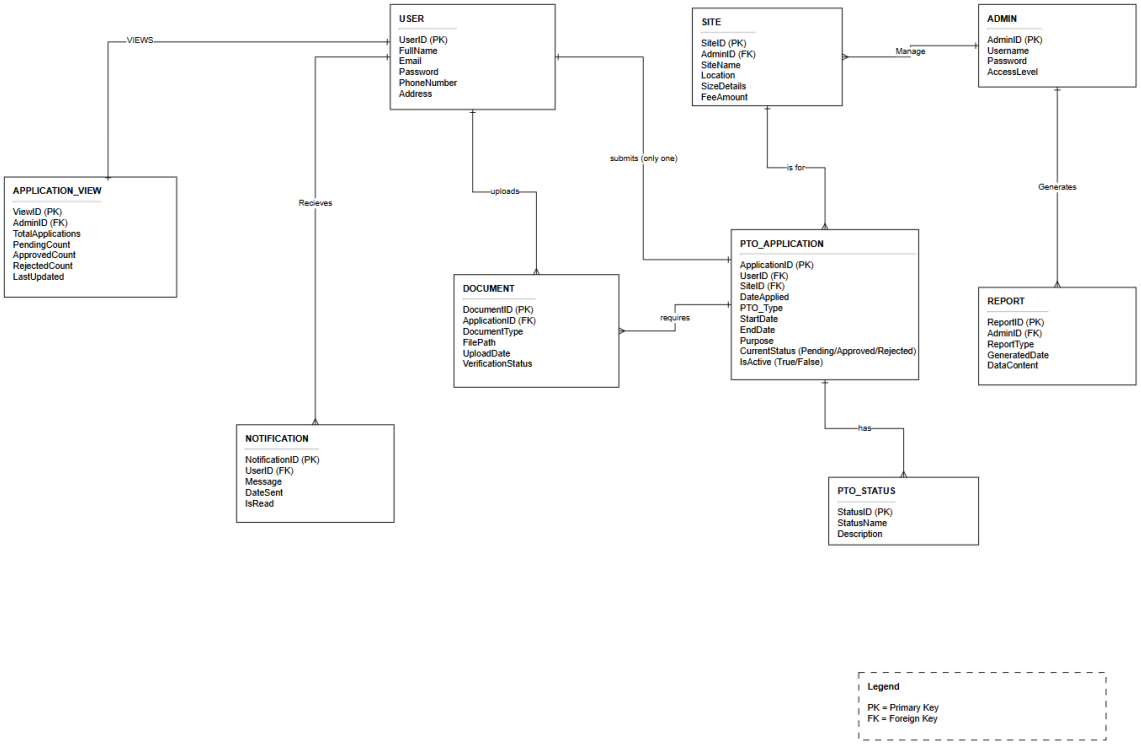


13.3 Database Design (ERD)

https://tinyurl.com/48anbe67

The Entity Relationship Diagram (ERD) illustrates the database structure of the Permission to Occupy (PTO) Management System and the relationships between its entities. It identifies eight main entities: User, Admin, Site, PTO_Application,

Document, PTO_Status, Notification, and Report, as well as an Application_View entity for displaying application statistics. The diagram demonstrates how users submit PTO applications, upload supporting documents, receive notifications, and track application statuses. Administrators manage sites, generate reports, and monitor applications. The relationships between entities are established using Primary Keys (PK) and Foreign Keys (FK), ensuring data integrity, consistency, and efficient management of PTO records within the Supabase database.



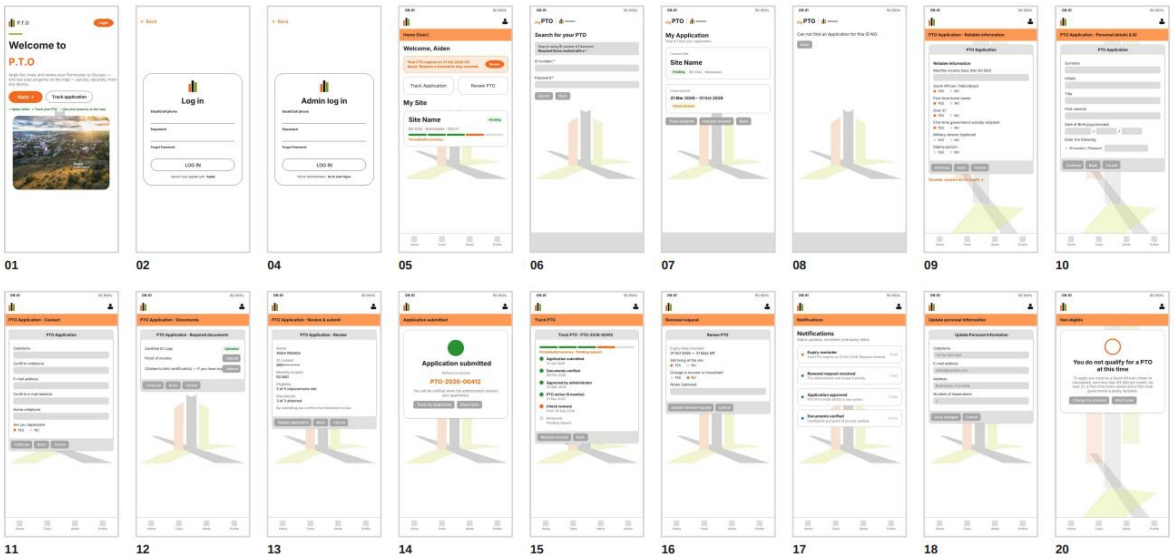
13.4 User Interface Wireframes

Mobile wireframe

https://www.figma.com/proto/hF3g70wiKh3NJvoP6Dmpen/Untitled?node-id=320-11925&p=f&t=ajQtHtzP4WYwDvLs-1&scaling=scale-down&content-scaling=fixed&page-id=300%3A689&starting-point-node-id=320%3A11925

Mobile Prototype

https://www.figma.com/design/hF3g70wiKh3NJvoP6Dmpen/Untitled?node-id=300-689&p=f&t=jQzNewh1hYUo0MM9-0

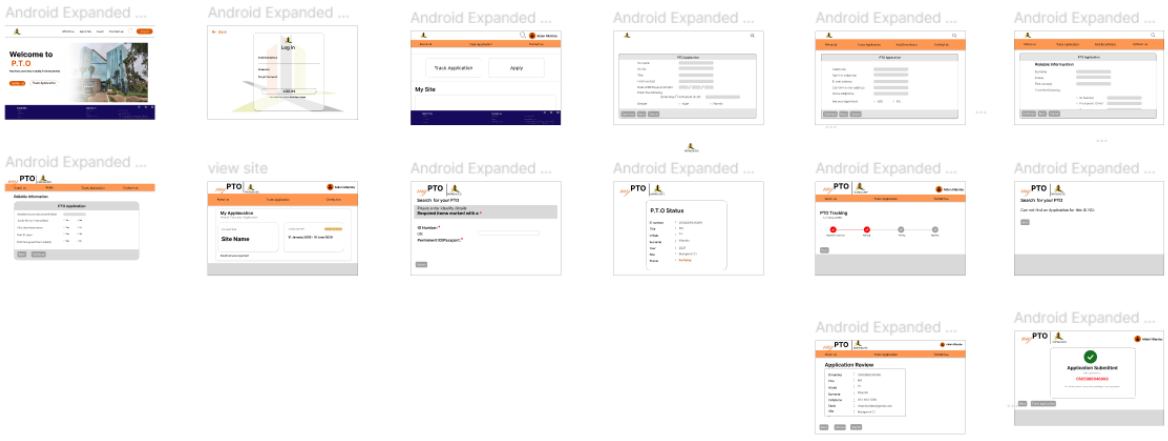


Web wireframe

https://www.figma.com/design/hF3g70wiKh3NJvoP6Dmpen/Untitled?node-id=0-1&p=f&t=jQzNewh1hYUo0MM9-0

web prototype

https://www.figma.com/proto/hF3g70wiKh3NJvoP6Dmpen/Untitled?node-id=106-16&t=f7781PNJRa1UwXdE-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=106%3A3



13.5 PTO Design Alignment

Requirement	Use Case	Sequence Diagram Flow	ERD Entity / Table	Interface / Screen
FR1 & FR16: User Authenticatio n & Access Control — Secure login and role-based access for system roles.	Login (includes Authenticate)	User / Officer / Admin → Flutter App / Web App → Supabase Authenticatio n	USER, ADMIN	Log in screen / Admin log in screen
FR3 & FR7: Beneficiary & Profile Management — Capture, view, and update beneficiary details.	Update Personal Information / View PTO Details & Status	Beneficiary / PTO Officer →Flutter App / Web App → Supabase Database	USER	PTO Application - Personal Details & ID screen, Contact Details screen, Update Personal Information screen
FR4: Site Management — Capture and manage available settlement site details.	Manage Sites (Add Sites / Remove Sites)	System Administrator Web Application → Supabase Database	SITE, ADMIN	Site Management screen / View Site screen

FR5: PTO Creation & Application Submission — Create PTO records and upload	**Submit PTO Application** (includes Upload documents)	Beneficiary / PTO Officer → Flutter App / Web App → Supabase Storage & Database	PTO_APPLICATION, DOCUMENT, SITE, USER	PTO Application - Required Documents screen, Review & Submit screen,
verification documents.				Application Submitted screen
FR6 & FR7: PTO Status Tracking — View and track PTO application status updates.	Track PTO Application status / View PTO Details & Status	Beneficiary / PTO Officer → Flutter App / Web App → Supabase Database	PTO_APPLICATION, PTO_STATUS	Search for your PTO screen, Track PTO / PTO Status screen
FR8 & FR9: Expiry Monitoring & Notifications — Monitor 6month PTO validity periods and dispatch expiry reminder alerts.	Receive Notification	Supabase Backend → External Notification Service → Flutter App / Web App → User	PTO_APPLICATION, NOTIFICATION	Notifications screen, Expiry alert reminders on Home Dashboard

FR10: PTO Renewal — Allow users/officials to submit and process renewal requests for expiring PTOs.	Submit Renewal Request	Beneficiary / PTO Officer → Flutter App / Web App → Supabase Database	PTO_APPLICATION, PTO_STATUS	Renewal Request screen, Home Dashboard (Renew PTO option)
FR11: PTO Deregistration — Record de-registration of invalid or	Manage PTO Records / View PTO Details & Status	PTO Officer / System Administrator → Web Application →	PTO_APPLICATION, PTO_STATUS	Administrativ e Web Portal — PTO Management screen
unoccupied PTO records.		Supabase Database		
FR12 & FR13: Search & Filtering System — Search and filter beneficiaries, sites, and PTO records by criteria.	Track PTO Application status / Search Records	User / Official → Flutter App / Web App → Supabase Database	PTO_APPLICATION, USER, SITE	Search for your PTO screen, Filter Records view
FR14: Dashboard & Municipal Reporting — Display overview metrics and compile system reports.	Generate Reports / View Dashboard	System Administrator → [suspicious link removed] Web Application → Supabase Database	REPORT, APPLICATION_VIE W, ADMIN	Admin Dashboard screen, Reports Filing screen

14. Conclusion

The proposed Permission to Occupy Management System will provide a digital platform for managing PTOs within informal settlements.

The primary problem identified by the stakeholder is the absence of a dedicated PTO management system and the current dependence on Excel spreadsheets and manual processes.

The proposed solution consists of a web application and mobile application supported by a database. The system will focus on beneficiary management, site management, PTO management, monitoring, verification, notifications and secure handling of personal information.

The initial implementation will use Flutter for the mobile application, C# and ASP.NET Core for the web application, Supabase for data storage and GitHub for source-code management.

Although the project is currently being developed as an academic project, the stakeholder expressed interest in potentially continuing with the system beyond the academic project. Any future municipal implementation will, however, be subject to the municipality's internal technology, security, procurement, licensing and governance processes.

15. Reference

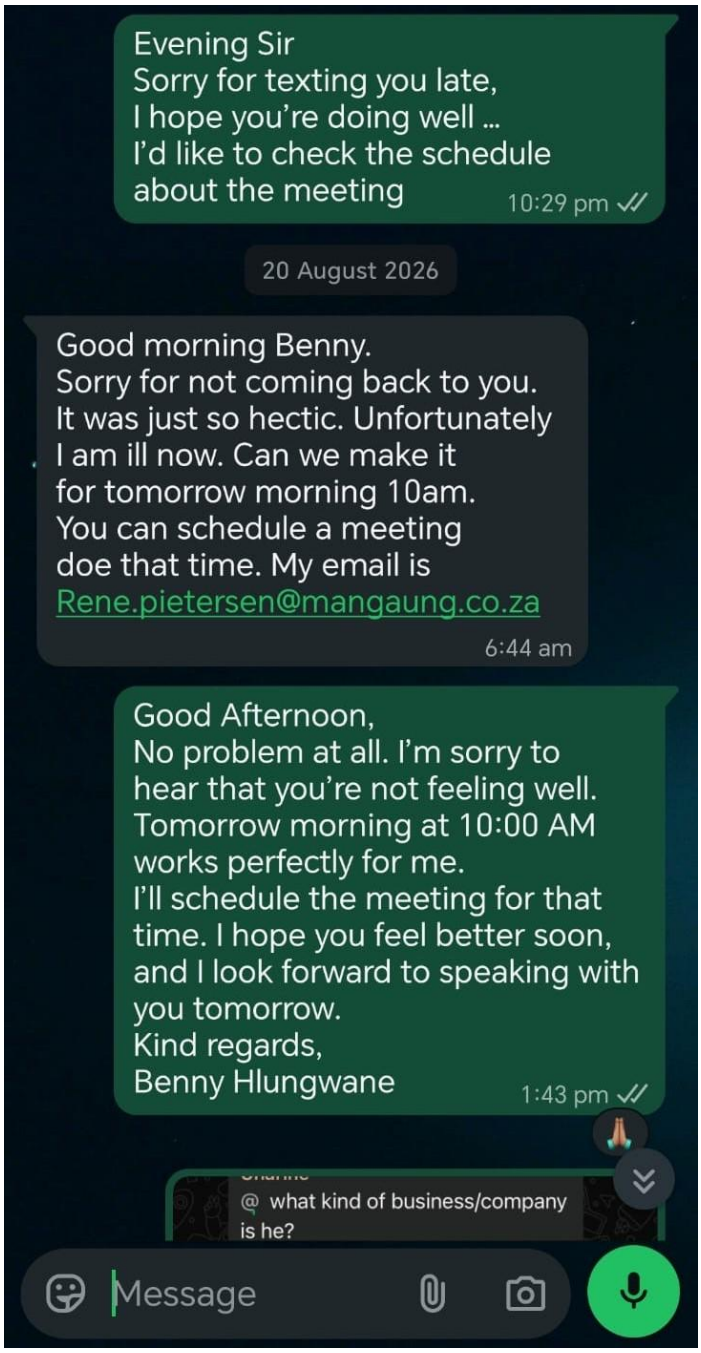
Stakeholders Sources

Mr Rene Pieterson, Mangaung Metropolitan Municipality Human Settlements Department. (2026). Stakeholders interview regarding the Permission to occupy (PTO) management process. Personal communication, 22 August 2026.

16. Appendences

Appendix A: Stakeholder Communication Evidence

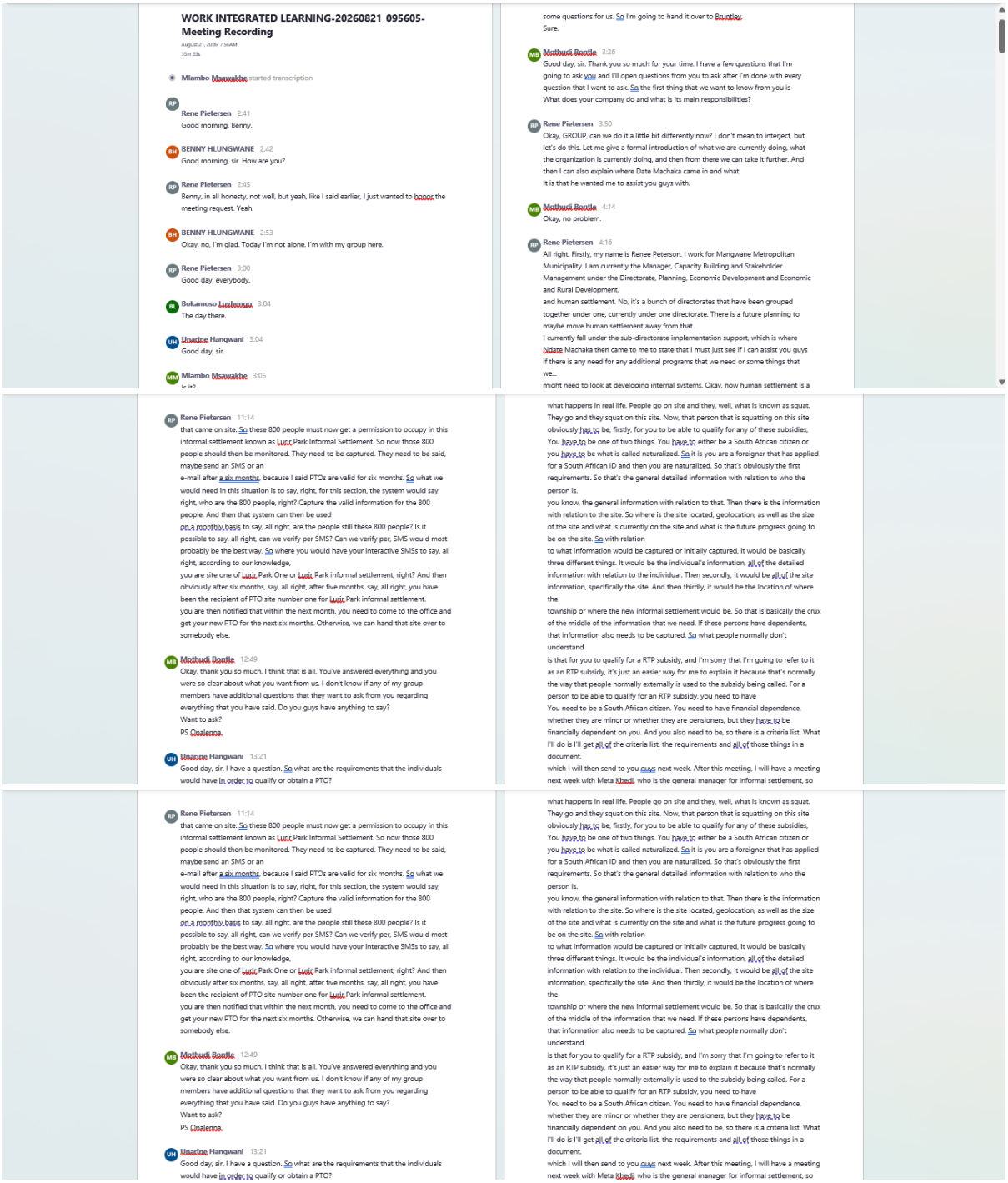
The following screenshot provide evidence of communication and engagement between the project group and the stakeholder regarding the proposed **PTO Management System**.



The screenshot was about confirmation of meeting

Appendix B: Stakeholder interview transcript 21/08/2026

The following screenshots provide evidence of our Microsoft Teams meeting with the stakeholder and form part of the transcript of the discussion.



WORK INTEGRATED LEARNING-20260821_095605-Meeting Recording

August 21, 2026, 7:56AM
35m 33s

Mlambo Msawakhe started transcription

Rene Pietersen 2:41
Good morning, Benny.

BENNY HLUNGWANE 2:42
Good morning, sir. How are you?

Rene Pietersen 2:45
Benny, in all honesty, not well, but yeah, like I said earlier, I just wanted to honor the meeting request. Yeah.

BENNY HLUNGWANE 2:53
Okay, no, I'm glad. Today I'm not alone. I'm with my group here.

Rene Pietersen 3:00
Good day, everybody.

Bokamoso Lunbengo 3:04
The day there.

Unarine Hangwani 3:04
Good day, sir.

Mlambo Msawakhe 3:05

some questions for us. So I'm going to hand it over to Bruntley.
Sure.

Mothudi Bontle 3:26
Good day, sir. Thank you so much for your time. I have a few questions that I'm going to ask you and I'll open questions from you to ask after I'm done with every question that I want to ask. So the first thing that we want to know from you is
What does your company do and what is its main responsibilities?

Rene Pietersen 3:50
Okay, GROUP, can we do it a little bit differently now? I don't mean to interject, but let's do this. Let me give a formal introduction of what we are currently doing, what the organization is currently doing, and then from there we can take it further. And then I can also explain where Date Machaka came in and what
It is that he wanted me to assist you guys with.

Mothudi Bontle 4:14
Okay, no problem.

Rene Pietersen 4:16
All right. Firstly, my name is Renee Peterson. I work for Mangwane Metropolitan Municipality. I am currently the Manager, Capacity Building and Stakeholder Management under the Directorate, Planning, Economic Development and Economic and Rural Development.
and human settlement. No, it's a bunch of directorates that have been grouped together under one, currently under one directorate. There is a future planning to maybe move human settlement away from that.
I currently fall under the sub-directorate implementation support, which is where Ndate Machaka then came to me to state that I must just see if I can assist you guys if there is any need for any additional programs that we need or some things that we...
might need to look at developing internal systems. Okay, now human settlement is a

Rene Pietersen 11:14
that came on site. So these 800 people must now get a permission to occupy in this informal settlement known as Lurir Park Informal Settlement. So now those 800 people should then be monitored. They need to be captured. They need to be said, maybe send an SMS or an
e-mail after a six months, because I said PTOs are valid for six months. So what we would need in this situation is to say, right, for this section, the system would say, right, who are the 800 people, right? Capture the valid information for the 800 people. And then that system can then be used
on a monthly basis to say, all right, are the people still these 800 people? Is it possible to say, all right, can we verify per SMS? Can we verify per, SMS would most probably be the best way. So where you would have your interactive SMSs to say, all right, according to our knowledge,
you are site one of Lurir Park One or Lurir Park informal settlement, right? And then obviously after six months, say, all right, after five months, say, all right, you have been the recipient of PTO site number one for Lurir Park informal settlement.
you are then notified that within the next month, you need to come to the office and get your new PTO for the next six months. Otherwise, we can hand that site over to somebody else.

Mothudi Bontle 12:49
Okay, thank you so much. I think that is all. You've answered everything and you were so clear about what you want from us. I don't know if any of my group members have additional questions that they want to ask from you regarding everything that you have said. Do you guys have anything to say?
Want to ask?
PS Ooalenna.

Unarine Hangwani 13:21
Good day, sir. I have a question. So what are the requirements that the individuals would have in order to qualify or obtain a PTO?

what happens in real life. People go on site and they, well, what is known as squat. They go and they squat on this site. Now, that person that is squatting on this site obviously has to be, firstly, for you to be able to qualify for any of these subsidies,
You have to be one of two things. You have to either be a South African citizen or you have to be what is called naturalized. So it is you are a foreigner that has applied for a South African ID and then you are naturalized. So that's obviously the first requirements. So that's the general detailed information with relation to who the person is.
you know, the general information with relation to that. Then there is the information with relation to the site. So where is the site located, geolocation, as well as the size of the site and what is currently on the site and what is the future progress going to be on the site. So with relation
to what information would be captured or initially captured, it would be basically three different things. It would be the individual's information, all of the detailed information with relation to the individual. Then secondly, it would be all of the site information, specifically the site. And then thirdly, it would be the location of where the
township or where the new informal settlement would be. So that is basically the crux of the middle of the information that we need. If these persons have dependents, that information also needs to be captured. So what people normally don't understand
is that for you to qualify for a RTP subsidy, and I'm sorry that I'm going to refer to it as an RTP subsidy, it's just an easier way for me to explain it because that's normally the way that people normally externally is used to the subsidy being called. For a person to be able to qualify for an RTP subsidy, you need to have
You need to be a South African citizen. You need to have financial dependence, whether they are minor or whether they are pensioners, but they have to be financially dependent on you. And you also need to be, so there is a criteria list. What I'll do is I'll get all of the criteria list, the requirements and all of those things in a document.
which I will then send to you guys next week. After this meeting, I will have a meeting next week with Meta Khedi, who is the general manager for informal settlement, so

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Appendix C: Project Risk Register

This appendix contains the detailed Project Risk Register for the Permission to Occupy (PTO) Management System.

Project Risk Register

Risk Score = Likelihood x Impact | 1-3 Low | 4-6 Medium | 7-9 High

Risk ID	Risk Statement	Category	Likelihood (1-3)	Impact (1-3)	Score	Level	Mitigation	Contingency	Risk Owner	Status
R07	SMS integration may not be available, affordable or technically compatible with the system before the project deadline.	Technical	2	2	4	Medium	Research the SMS integration requirements early and design the notification module so that the SMS provider can be changed.	Use simulated notifications during the academic prototype and implement the actual SMS service when the required provider is confirmed.	Integration Developer	Open
R08	The proposed Supabase environment may not meet the municipality's future hosting, security, licensing or infrastructure requirements.	Technical	1	2	2	Low	Design the system with separation between the application and database layers and document the database structure for possible migration.	Deploy the academic prototype using Supabase while preparing the system and database documentation for future migration to municipal infrastructure.	Technical Lead	Open
R09	The system may not scale effectively if large numbers of beneficiaries and sites are captured, such as the stakeholder's example of approximately 800 beneficiaries in one settlement.	Technical	2	2	4	Medium	Use an appropriate database structure, indexed searches and efficient queries. Test the system using realistic amounts of data.	Optimise database queries and system resources or restructure affected components before deployment.	Backend Developer	Open
R10	Incorrect beneficiary, site or PTO information entered into the system could lead to inaccurate monitoring and incorrect administrative decisions.	Quality	2	2	4	Medium	Use input validation, required fields, confirmation screens and appropriate data formats for ID numbers, dates and site information.	Allow authorised users to correct records and maintain an audit/history of important changes.	Database Developer / Adm	Open
R11	The project team may not complete all planned functionality within the academic Phase One deadline.	Schedule	3	2	6	Medium	Prioritise core PTO functionality, assign tasks to team members and monitor progress against milestones.	Deliver the minimum viable system containing the highest-priority requirements first and defer lower-priority features to a later phase.	Project Manager	Open
R12	Future migration from the student-developed system to municipal infrastructure may require changes because of municipal licensing, technology and governance requirements.	Technical	1	2	2	Low	Follow standard development practices, document the architecture and database, and avoid unnecessary dependency on project-specific infrastructure.	Treat the academic system as a prototype and conduct a separate migration phase if the municipality approves continued development.	Technical Lead / Stakehold	Open

Note: Likelihood and Impact ratings were reassessed by the project team using the 1-3 scale in Activity 2.2, consistent with the priority order identified during stakeholder analysis.

Highest Risk	Why Is This Important?	What Must the Group Do Now?
1. R01 - Delayed or Incomplete Stakeholder Requirements	This is currently the highest project risk. The stakeholder indicated that additional information is still required regarding the PTO process, documentation, front-end requirements and individual beneficiary information, and the complete eligibility criteria have not yet been confirmed.	Continue requesting the outstanding requirements from the stakeholder, document all confirmed requirements as the baseline, clearly mark outstanding items as TBD, and proceed with development using the confirmed requirements.
2. R02 - Municipal Approval and Internal Processes	The municipality has internal processes that must be followed before the project can be adopted or continued within the organisation, involving stakeholders such as the Chief Technology Officer. Municipal licensing, governance and public-accountability requirements could prevent the municipality from simply adopting the project's freeware solution.	Treat municipal deployment as a separate future phase while continuing to develop the academic prototype using the confirmed requirements, and identify the municipal stakeholders who must approve or validate the system.
3. R11 - Failure to Complete the Project Within the Academic Deadline	The team has a limited academic deadline for Phase One, due by the end of August. The project also depends on information that may only arrive after development has already started.	Prioritise core PTO functionality and divide development tasks among team members. Complete the Minimum Viable Product (MVP) first and defer lower-priority functionality to a later development phase.

Overall Project Risk: HIGH - two of the highest risks (R01, R02) relate to stakeholder requirements and municipal processes and should receive immediate attention, as they affect requirements, design and future deployment of the system.

Risk Register | Rating Scales | Highest Risks Review | +

Risk Rating Scales

Likelihood

Rating	Likelihood	Meaning
1	Unlikely	The risk is not expected to happen
2	Possible	The risk could happen
3	Likely	There is a strong possibility that the risk will happen

Impact

Rating	Impact	Meaning
1	Low	Small effect on the project
2	Medium	Noticeable effect requiring action
3	High	Serious effect on project completion or quality

Risk Score = Likelihood x Impact

Risk Score	Risk Level
1 - 3	Low
4 - 6	Medium
7 - 9	High

High risks require particular attention from the group.

Risk Register | Rating Scales | Highest Risks Review

Project Risk Register

Risk Score = Likelihood x Impact | 1-3 Low | 4-6 Medium | 7-9 High

Risk ID	Risk Statement	Category	Likelihood (1-3)	Impact (1-3)	Score	Level	Mitigation	Contingency	Risk Owner	Status
R01	Required PTO criteria, documentation and business rules may not be received from the stakeholder in time, resulting in incomplete or incorrect system requirements.	Stakeholder	3	3	9	High	Maintain regular communication with the stakeholder and document all confirmed requirements. Use the stakeholder meeting as the baseline and clearly mark	Develop the system using confirmed requirements and update the SRS and system when the outstanding information is received.	Project Manager / Stakeho	Open
R02	Municipal internal approval processes may delay confirmation of requirements, access to information, or future adoption of the system.	Stakeholder	3	3	9	High	Identify required municipal stakeholders early and document dependencies and approvals.	Continue the project as an academic prototype while keeping the system designed for possible future municipal deployment.	Project Manager / Stakeho	Open
R03	Sensitive beneficiary information such as ID numbers and personal details could be exposed or accessed by unauthorised users.	Data and security	2	2	4	Medium	Implement authentication, role-based access, secure database access and restrict sensitive information to authorised users.	Remove/restrict exposed information, investigate the incident, change credentials and restore the system from a secure backup where necessary.	Backend Developer / Secu	Open
R04	Limited project funding may restrict the team's ability to use paid hosting, database services, SMS services or other required infrastructure.	Resources	2	2	4	Medium	Use cost-effective technologies such as Supabase and free/low-cost development resources during the academic phase. Monitor service limits and costs.	Reduce non-essential functionality or use alternative approved services until funding or municipal infrastructure becomes available.	Project Manager / Technic	Open
R05	The current PTO process is largely manual and spreadsheet-based, which may make it difficult for the team to accurately translate existing processes into system requirements.	Quality	2	2	4	Medium	Document the current PTO process and validate workflows with the stakeholder before implementation.	Implement the confirmed core workflow first and revise additional workflows after stakeholder clarification.	Business Analyst / Stakeh	Open
R06	Incorrect implementation of the six-month PTO validity and expiry process could result in beneficiaries receiving incorrect expiry dates or notifications.	Quality	2	2	4	Medium	Define and test PTO start-date, expiry-date, renewal and notification rules before implementation.	Correct affected records manually and issue corrected notifications while the system logic is fixed.	Backend Developer / Teste	Open
R07	SMS integration may not be available, affordable or technically compatible with the system before the project deadline.	Technical	2	2	4	Medium	Research the SMS integration requirements early and design the notification module so that the SMS	Use simulated notifications during the academic prototype and implement the actual SMS service when the required	Integration Developer	Open

Appendix D: Stakeholder Interview Transcript 17/09/2026

The following screenshots provide evidence of our Microsoft Teams meeting with the stakeholder and form part of the transcript of the discussion.

WIL-20260917_081339-Meeting Recording

September 17, 2026, 6:13AM
18m 9s

Mlambo Msawakhe started transcription

Hangwani Unarine 0:04
Good morning once again, sir, and thank you for giving us time. So we are pleased to inform you that our presentation last week went wrong, and our lectures are really looking forward to seeing the final product. So
Yeah, our phase one went close, so now we are on our phase two, which is the designing phase. So, as Benny's message said, we need more guidance on how we should approach the whole solution. So, Wuntly is going to take you to our proposed solution, and then you can just let us.
Or if you are happy with it, is there a certain thing that we should change, or are we approaching the solution in the correct manner?

Rene Pietersen 1:04
All right, no, you guys can go ahead and then I can give guidance as to that.

Mothudi Bontle 1:10
Okay, good morning, sir. I hope you're good. Thank you so much for your time. So based on the information of the problem that you stated the last time we had a meeting, we came up with a solution that our system will have two actors, which is the user
the user, which is the beneficiary, as well as the administrator slash PTO officer. So the beneficiary will be able to log in, apply for the PTO, view PTO records, view the expiring date, view the PTO status.
Receive expiry notifications, request PTO renewal, review, view renewal status, view the profile, manage notification, log out, and then the administrator/PTO officer will. Be able to manage users, manage records, manage sites, manage...
Manage site, view PTO records, generate reports, track PTO, view expiry PTOs, update PTO statuses, renew PTO, deregister PTO, create a PTO or approve the PTO from the information that was given.
from the beneficiary. Like they're going to look at the beneficiary information and from that information, approve the PTO. And also, regarding the requirements that the beneficiary has to have, you only mentioned 2, so
And you said you're going to send us like more information, so we only know like 2... requirement that the beneficiary has to have in order to qualify for the PTO. So yeah, that is that. So thank you.

Rene Pietersen 3:13
All right, no, everything there seems in order. Guys, I'm going to be very honest with you and I'm going to be brutally honest and it's not going to be nice. But I need to be upfront with you so that you understand where we are with the process. Right now, I'm getting a lot of
being sent around. Look at the PTO processes, it's not my process, okay? It's a process that is being done by informal settlement. I have contacted several managers, however, I've not managed to be able to get any information out of them. There does not seem to be any forms.
or anything like that. Within the current situation that the municipality finds itself, it is, I think it's like 49 days or something like that before elections. So this is a crucial time for the people to go out to do things. Our programs are running frantically. Right now, my team is out. They are, I'm not even going to see them though, because the whole week, this week, they are out with them.
the master's office and the whole week next week, they are out with the premier and the speaker's office. Once again, I will try to get information. If I do not manage to get the information, I will let Benny know, but I'm going to try to get information. The reason why I've not been able to get back to you is because I don't get the information.
That you might need.
Subsequent to the meeting that I had with Meta Khedi as well and with the CTO, which I told Benny about, it also seems that the National Department of Human Settlement has now identified PTOs as a problem within the country and the National Department of Human Settlement
is also in the process of developing more or less a similar programme. I was not aware of that when we had our initial meeting. It is only after I had a meeting with Meta Kheri where she stated that according to them, the National Department of Human Settlement is busy with that programme and if the National Department of Human Settlement is running that programme, it would be better because
It would also come at no cost to the municipality.
So bearing those two things in mind, I needed to be upfront with you to tell you that that's the process. Hence, my last conversation with Benny, I stated that this will be a desktop exercise. As much as your lecturers are excited about the prospect of the program, because of the limitations that we have internally,
That is where we are. Am I clear? Do you guys understand that? And then how does that affect your program going forward?

Hangwani Unarine 5:59
Sir, I think we are. I think it's clear enough. Well, so, so we don't mind, as we say, this is an academic project, so it doesn't really need to work after everything. So, I think I think what we have so far is enough for us to continue with our...
Academic part of the project.

Rene Pietersen 6:23
Okay, once again, like I said, I've reached out to two different managers, which then once again, they want to go into the process of having a meeting with you guys, having this, having those processes, to which I explained to them, but it does not help. All that you guys want is you want the information. Okay, so you want to know what is the process and
For some other reason, they find it very difficult to discuss the process with me, which in all honesty, I don't understand. If the process is there and this is a standard process, just explain to me the standard process so that I can come back and give the information through to you. What I'm going to do is I'm going to reach a little bit lower.
I'm going to speak to somebody else.
But when do you need it, guys? Because it's a bit tough on my side. I've got a meeting. That's why I asked Benny to book this meeting so early today, because I've got a meeting scheduled from 9 for the rest of the day with relation to HR skills development, which I have to be part of, because the panel has to have consensus. And then tomorrow I am leaving to Johannesburg. So on my side, it's a bit tough this week. I don't think I'm going to be able to get anything done this week. Maybe Monday or Tuesday I can go through to see people that are a little bit lower in the process and maybe they can explain the process to me.

Hangwani Unarine 7:52
Bontle
Would you want to answer?

Mothudi Bontle 7:57
Yeah, I was going to say we have, like, we have enough information to continue with the project. The only thing that we wanted to know was the requirements, but we now have two. So I think if you won't be able to gather information or if they won't be able to tell you,
More about the requirements, we'll just use the ones that we have, the two that we have.

Rene Pietersen 8:23
Yeah, look, it's, and something, remember, it's informal settlement, okay? And it's in the name. So it is when people go and they say, all right, fine, here is an open piece of land. I'm going to, I'm going to send Benny something now. I just sent an open piece of land, for example, now.
through to my GM. It's just something that, it's on TikTok, but I mean, this is an open piece of land. So what normally would happen is this piece of land would then be used for informal settlement. So people would then go and they will erect shacks, not in a right way, not having
sites or anything like that, then that informal settlement has to be formalized. So it's just that permission to occupy is just those couple of criteria because you cannot actually go and say, all right, you now need all of these requirements to be able to be on the site.
It does not work that way in reality. In practice, it might work that way, but in reality, normally what would happen is people will go, they will, in the old way of putting it, they will go and squat. And once they have squatted, there's nothing that you can do with relation to removing them. The Constitution does not then allow you to say, right,
Scrap this old place. I don't know, you guys probably know the red ants, am I right?

Rene Pietersen 9:52
That's where the red ants come in, because now they would go to a piece of land

It would also come at no cost to the municipality.
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Scrap this old place. I don't know, you guys probably know the red ants, am I right?

Rene Pietersen 9:52
That's where the red ants come in, because now they would go to a piece of land that is not suitable for human accommodation. Then they need to be removed from that land.

Rene Pietersen 10:05
then that's where the red ends come in, because now it's a forceful removal where you have a court order. So the default things that I've given you, that's basically the minimum requirements. It's unlike with relation to the RDP process where you have all of these various

Rene Pietersen 10:25
requirements that you have to follow. And if you don't adhere to one of the requirements, then you are kicked out. It's not the same process, if that makes sense.

Mothudi Bontle 10:37
Okay, so I now understand. Maybe we didn't truly understand. I don't know. We thought maybe in order for a person to qualify for that, they have to have certain requirements.

Rene Pietersen 10:57
No, that's the thing. It's not, it's not a, hence I'm saying it's an informal process. The same as with relation to informal settlement, it is an informal process. What then should happen as the process progresses, okay, you have this informal piece of land. The informal piece of land has to be formalized. So now what you do is you are then saying, all right.
If we handle a clean piece of land, the process is supposed to work as follows. People come and they settle on the land informally so. The next step would be for the council to recognize them and then to go to the municipality first has to approve it. Then they have to go through a process of formalizing basically the township, let's call it that. Okay. Then sites have to be measured out. Now there are regulations within the municipality as to what the size of the sites must be, where water must come on. So then that process starts to be formalized. But
When the people are there and then the initial sites are pinned out, that's when your PTO gets issued out.
After that, once your PTO has been issued out, the process should be to formalize the informal settlement. That's another process that they have to go through to the land meter general and all of those things. Now that that process has been formalized, now the township is then named, the sites are then recognized.
What should an employee is a person should then get a house built through another human settlement process. Let's call it RDP. I know the name has changed, but for all purposes, then the person gets a subsidy. Now, there's a list of criteria that he needs to adhere to to be able to qualify for the subsidy.
But in a perfect situation, let's handle this as a perfect situation. Those people then get access to RDP housing subsidy, which means that the site is now formalized and through the RDP subsidy, a house is built.
And then after the house has been built, then a person gets a title deed that gets issued to you.
So it's a very, very, very lengthy process. The PTO is just the first part of that whole lengthy process. Does that make sense?

Hlungwane Benny 13:27
Yes, Sir.

Rene Pietersen 13:29
All right. Okay. So guys, what I like I said, I'll try to reach out to the guys Monday. Benny, I'll give you Monday, Tuesday. I'll give you feedback via WhatsApp. Let's say Tuesday, by Tuesday, the latest. If I unfortunately, if I don't get a hold of anybody that can explain that.
actual processes on the ground, you guys continue as is, please. I don't want to be the stumbling block for your processes to go on. But unfortunately, if I don't get anything else from the colleagues, then I would like you guys just to continue with your processes so that your project can start running.
Oh, continue, sorry.

Hlungwane Benny 14:13
No, it's fine. Thank you very much, sir. I think all Naledi wanted to know was the admin, is he the one who's supposed to create an application for the resident, or maybe the residents create an application for themselves?

Rene Pietersen 14:31
Okay.

Hlungwane Benny 14:31
Like, is the system for the municipality or for everyone?

Rene Pietersen 14:37
Look, the system in this scenario, the system would need to be for the user within humans, within informal settlement. So you would have, theoretically speaking, you would have a systems administrator, okay? And then you would have a user.

Rene Pietersen 14:56
and then you would have a beneficiary.
So your systems administrator needs to be the one that assigns a user.
and then the user should be able to log into the system and capture both the beneficiary and the site details.

Hlungwane Benny 15:16
Ohh, okay.

Rene Pietersen 15:20
And then the beneficiary should be able to have viewing details with relation to what their status would be on the application.

Hlungwane Benny 15:33
Okay.

Rene Pietersen 15:36
So that for me would be the ideal where you would have a systems administrator, okay, which theoretically speaking, let's say that's the manager of the unit.

Hlungwane Benny 15:43
Yeah.

Rene Pietersen 15:50
which in certain cases would be me Pakane or me Mie for Tebantu. So the person for Tebantu, theoretically speaking, would then register the users on the system, which is the people that can capture both the beneficiary and the site information.
And then at the end of that, the beneficiary has to have access to the system, but obviously the beneficiary would then only have viewing rights to the system.
with relation to, okay, yes, I've got a PTO. The PTO is I, Renee Peterson, ID number, cell phone number, site number, PTO status.
So if I'm the beneficiary, I then say, all right, I want to see on the system, if I am on the system for this ward, then yes, I am then able to view the system and that's it.

Hlungwane Benny 16:55
I think we all get it. Thank you very much, sir.

Rene Pietersen 16:58
All right.
Any other questions, guys?

Hlungwane Benny 17:12
I think we are sorted to say. Thank you for your time.

Rene Pietersen 17:15
Alright, alright, pleasure guys. Good luck.

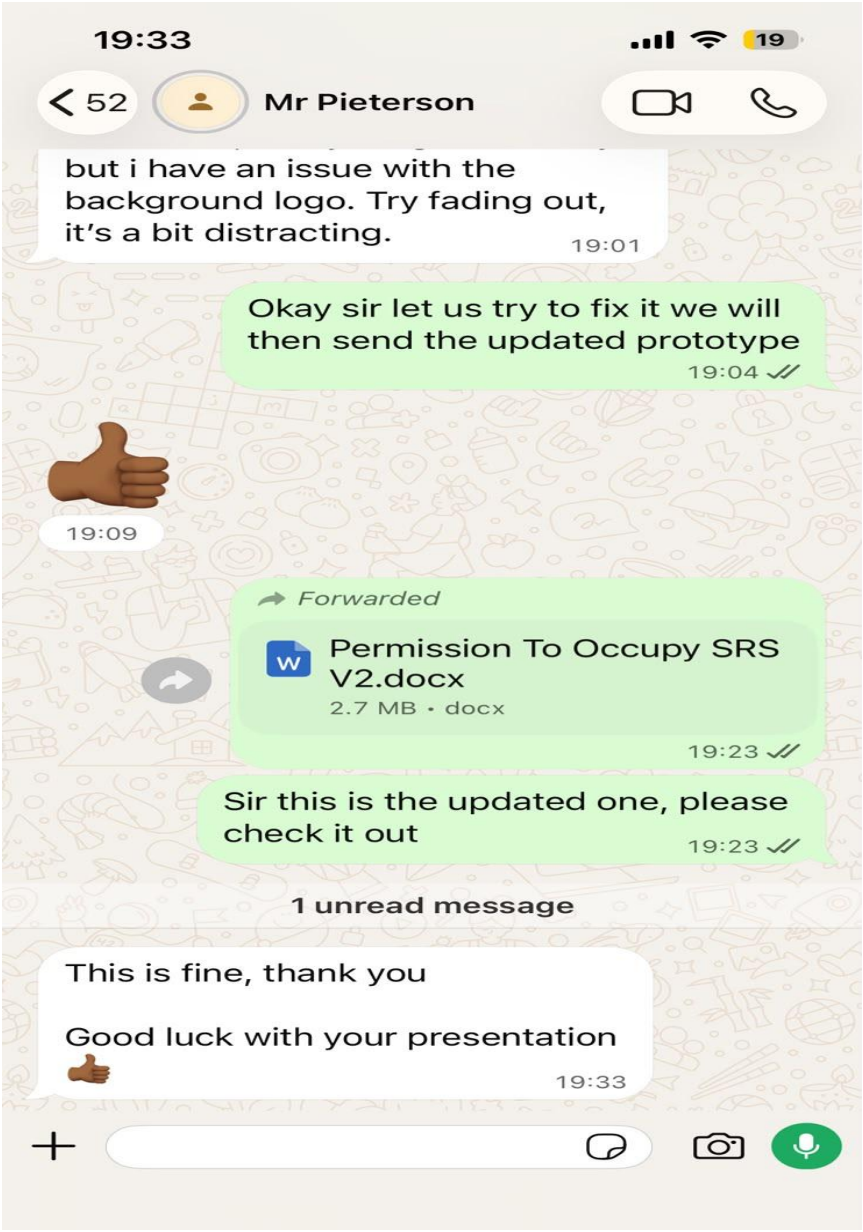
Mlambo Msawakhe 17:20
Thank you, thank you, sir.

Rene Pietersen 17:22
Thanks, take care.

Mlambo Msawakhe stopped recording

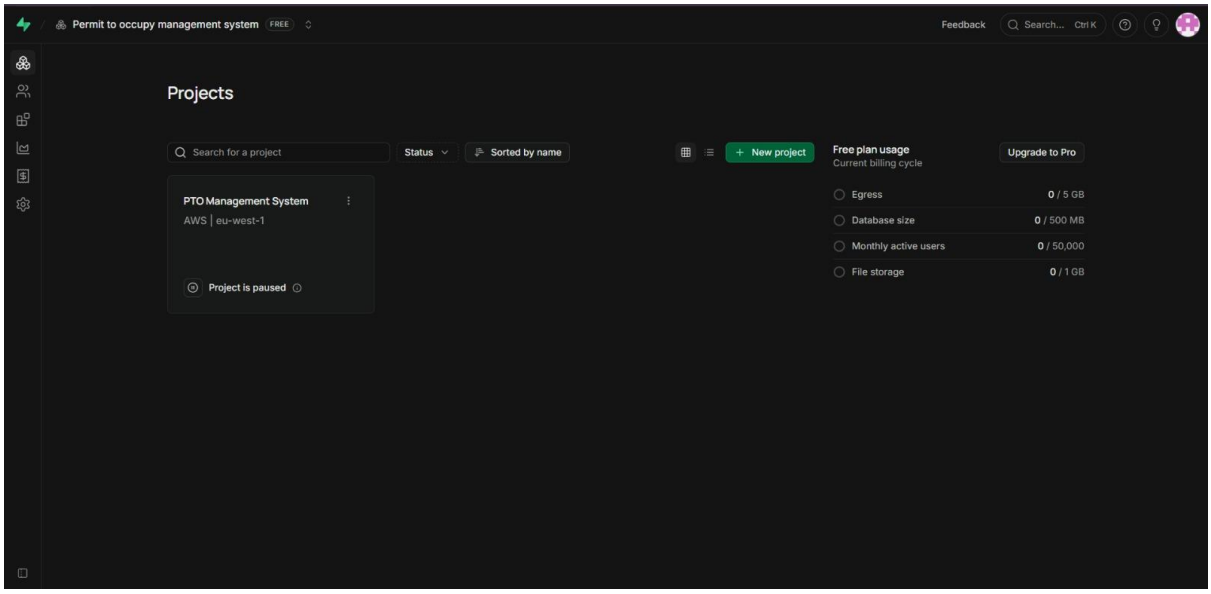
Appendix D: Stakeholder feedback

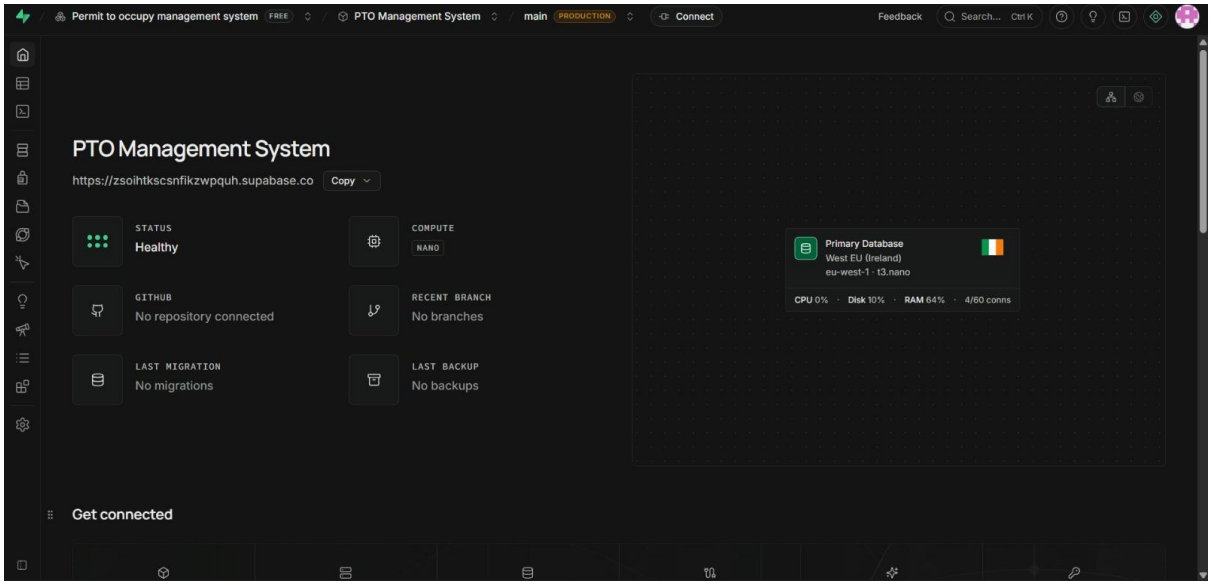




Appendix E: Supabase Database Setup Evidence

The following screenshots provide evidence of the **Supabase environment created for the PTO Management System**. Supabase will be used as the database component of the system to store and manage PTO-related information and support the integration between the Flutter mobile application and the ASP.NET MVC web application.





17. GitHub Link:

https://github.com/HlungwaneB/PTO-Management-System